multi_agent_financial_analysis

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1 Multi-Agent Financial Analysis System

AAI-520 Final Team Project — Brandon Wirgau, Christina, and Isaac

This notebook is the canonical, executable report for an agentic investment-research system. It accepts a stock ticker and user research question, uses LangGraph to route a bounded run to market, financial, news, or comprehensive specialists, evaluates and revises the complete report, and stores reusable lessons for later runs.

Repository: AAI-520 Final Project

> Research support only. The output is not personalized investment advice and does not guarantee future performance.

1.1 1. Project objectives and implemented capabilities

The project demonstrates four autonomous agent functions:

1. **Plan** research objectives, questions, and tool selection using prior-run memory as context.
2. **Use tools** to retrieve Yahoo Finance market/fundamental data, SEC EDGAR facts and filings, and multi-source company news.
3. **Reflect and evaluate** data quality, report grounding, and specialist-analysis completeness.
4. **Learn across runs** by persisting bounded, timestamped research lessons.

It also implements all three required workflow patterns:

- **Prompt chaining:** ingest → preprocess → classify → extract → summarize.
- **Routing:** the user request selects market-only, financial-only, news-only, or comprehensive research; news articles also route to earnings, market, and general-news specialists.
- **Evaluator–optimizer:** article analysis uses a four-part rubric, while the complete report is scored for completeness, grounding, reasoning, consistency, sources, and clarity and revised through a bounded LangGraph loop.

The refactored modules preserve the original notebook's planning, tool execution, validation, reflection, reporting, memory, SEC EDGAR retrieval, 10-year price graph, multi-source news analysis, specialist routing, and iterative optimization. The updated Google News RSS fallback and retry-aware HTTP behavior are included as well.

1.2 2. Implemented system architecture

The research workflow keeps provider adapters, provider-independent tools, agents, workflows, memory, and presentation concerns separate. This makes each stage inspectable and independently

testable.

```
flowchart LR
    U[User ticker and research question] --> P[Planner]
    L[Research memory] --> P
    P --> RQ{Request route}
    RQ -->|Market| MA[Market specialist]
    RQ -->|Financial| FA[Financial specialist]
    RQ -->|News| C[News prompt chain]
    RQ -->|Comprehensive| MA
    RQ -->|Comprehensive| FA
    RQ -->|Comprehensive| C
    Y[Yahoo Finance] --> MA
    Y --> FA
    S[SEC EDGAR] --> FA
    Y --> C
    N[NewsAPI] --> C
    G[Google News RSS fallback] --> C
    C --> R[Specialist router]
    R --> E[Earnings specialist]
    R --> M[Market specialist]
    R --> W[News specialist]
    E --> V[Evaluator / optimizer]
    M --> V
    W --> V
    MA --> Z[Synthesizer]
    FA --> Z
    V --> Z
    Z --> Q{Six-part report evaluator}
    Q -->|Below threshold| O[Report optimizer]
    O --> Q
    Q -->|Pass or revision limit| F[Final report and visuals]
    F --> L
```

1.3 3. Environment setup

Create and activate a virtual environment, install `requirements.txt`, start Ollama, and copy `.env.example` to `.env`. At minimum:

- `OLLAMA_MODEL=llama3.2` and a running `ollama serve` process;
- `SEC_USER_AGENT` containing an application name and monitored contact email;
- optional `NEWS_API_KEY` (Yahoo remains primary and Google RSS is a fallback);
- optional `DEFAULT_TICKER`, `RESEARCH_REQUEST`, `MAX_NEWS_ARTICLES`, `EVALUATION_SCORE_THRESHOLD`, `MAX_OPTIMIZER_ITERATIONS`, `REPORT_SCORE_THRESHOLD`, and `MAX_REPORT_REVISIONS`.

Credentials are read only from environment variables and are never displayed below.

[1]:
```
import getpass
import os
import shutil
import subprocess
import sys
import time
import urllib.request
from pathlib import Path

try:
    import google.colab  # noqa: F401

    IN_COLAB = True
except ImportError:
    IN_COLAB = False

if IN_COLAB:
    repo_url = "https://github.com/bwirgau-usd/
 ↪AAI-520-final-project-multi-agent-financial-analysis.git"
    repo_dir = Path("/content/
 ↪AAI-520-final-project-multi-agent-financial-analysis")
    if not repo_dir.exists():
        subprocess.run(
            ["git", "clone", "--branch", "main", repo_url, str(repo_dir)],
            check=True,
        )
    os.chdir(repo_dir)
    subprocess.run(
        [sys.executable, "-m", "pip", "install", "-q", "-r", "requirements.
 ↪txt"],
        check=True,
    )

    if not os.getenv("NEWS_API_KEY"):
        news_key = getpass.getpass(
            "NEWS_API_KEY (optional; input hidden): "
        ).strip()
        if news_key:
            os.environ["NEWS_API_KEY"] = news_key
    if not os.getenv("SEC_USER_AGENT"):
        os.environ["SEC_USER_AGENT"] = input(
            "SEC user agent (name and contact email): "
        ).strip()

    if shutil.which("ollama") is None:
        subprocess.run(
            ["apt-get", "install", "-y", "-qq", "zstd"],
```

```
            check=True,
        )
        installer = urllib.request.urlopen(
            "https://ollama.com/install.sh",
            timeout=60,
        ).read()
        subprocess.run(["sh"], input=installer, check=True)
    subprocess.Popen(
        ["ollama", "serve"],
        stdout=subprocess.DEVNULL,
        stderr=subprocess.DEVNULL,
    )
    time.sleep(5)
    subprocess.run(
        ["ollama", "pull", os.getenv("OLLAMA_MODEL", "llama3.2")],
        check=True,
    )
    print("Colab environment ready.")
else:
    print("Local environment detected; start Ollama separately with ollama␣
 ↪serve.")
```

```
Local environment detected; start Ollama separately with ollama serve.
```

[2]:
```
import os
import sys
from pathlib import Path

import pandas as pd
from IPython.display import JSON, Markdown, display
from dotenv import load_dotenv


def find_project_root(start: Path) -> Path:
    """Find the repository root from JupyterLab or a nested notebook path."""
    for candidate in (start.resolve(), *start.resolve().parents):
        if (candidate / "src").is_dir() and (candidate / "requirements.txt").
 ↪is_file():
            return candidate
    raise FileNotFoundError("Could not locate the project root.")


PROJECT_ROOT = find_project_root(Path.cwd())
if str(PROJECT_ROOT) not in sys.path:
    sys.path.insert(0, str(PROJECT_ROOT))
load_dotenv(PROJECT_ROOT / ".env")
```

```
from src.data_sources.yahoo_finance import YahooFinanceClient
from src.reporting import (
    plot_10_year_stock_price,
    plot_news_analysis_summary,
    plot_news_workflow,
    plot_system_architecture,
    render_console_summary,
)
from src.tools.registry import TOOLS
from src.workflows import build_research_workflow

TICKER = os.getenv("DEFAULT_TICKER", "AAPL").strip().upper()
RESEARCH_REQUEST = os.getenv(
    "RESEARCH_REQUEST",
    "How do market performance, financial condition, SEC evidence, and recent␣
 ↪news affect the company's current risk picture?",
).strip()
print(f"Project root: {PROJECT_ROOT}")
print(f"Ticker: {TICKER}")
print(f"Research request: {RESEARCH_REQUEST}")
print(f"NewsAPI configured: {bool(os.getenv('NEWS_API_KEY'))}")
print(f"SEC identity configured: {bool(os.getenv('SEC_USER_AGENT'))}")
```

```
/Users/brandonwirgau/.local/share/virtualenvs/brandonwirgau-
gZAqjFTS/lib/python3.13/site-packages/requests/__init__.py:113:
RequestsDependencyWarning: urllib3 (2.7.0) or chardet (7.4.3)/charset_normalizer
(3.4.4) doesn't match a supported version!
  warnings.warn(
```

```
Project root: /Users/brandonwirgau/Projects/AAI/520/Final Project
Ticker: AAPL
Research request: How do market performance, financial condition, SEC evidence,
and recent news affect the company's current risk picture?
NewsAPI configured: False
SEC identity configured: False
```

```
/Users/brandonwirgau/.local/share/virtualenvs/brandonwirgau-
gZAqjFTS/lib/python3.13/site-packages/langgraph/cache/base/__init__.py:8:
LangChainPendingDeprecationWarning: The default value of `allowed_objects` will
change in a future version. Pass an explicit value (e.g.,
allowed_objects='messages' or allowed_objects='core') to suppress this warning.
  from langgraph.checkpoint.serde.jsonplus import JsonPlusSerializer
```

1.4 4. Agent design, data sources, and tool use

Component	Implemented responsibility
Planner Agent	Uses the ticker, registered tools, and the last three relevant memories to produce objectives, questions, and an allow-listed tool plan.
Tool Executor	Runs only registered tools and isolates provider errors so one failure does not terminate the research run.
Market Specialist / Tools	Retrieve price and volume history; compute one-month, three-month, and one-year returns, volatility, drawdown, 52-week range, and average volume.
Financial Specialist / Tools	Retrieve income, cash-flow, and balance-sheet statements; compute revenue/earnings growth and analyze cash, debt, net debt, assets, liabilities, and equity.
SEC EDGAR Tool	Resolves CIK, reads official company facts, and lists recent 10-K, 10-Q, and 8-K filings with source metadata.
News Prompt Chain	Combines Yahoo and NewsAPI; invokes Google RSS when primary coverage is missing or sparse; deduplicates and structures articles.
Specialist Router	Sends classified items to earnings, market, or general-news specialists.
Evaluator / Optimizer	Scores observations, interpretation, uncertainty, and evidence breadth; revises until the threshold or iteration limit.
Synthesis Agent	Produces a grounded Markdown report spanning market, financial, filing, and news evidence.
Memory Curator / Store	Saves bounded, timestamped lessons using atomic file replacement; memory is planning context, never current evidence.

FRED and Alpha Vantage packages remain deliberate extension points. The report does not claim they are integrated data sources.

[3]:
```
tool_rows = [
    {
        "tool": name,
        "role": {
            "price_data": "1m/3m/1y performance, risk, range, and volume",
            "company_info": "Profile, valuation, margins, and dividend",
            "financials": "Income-statement trends",
            "cash_flow": "Operating cash flow, free cash flow, and capex",
            "balance_sheet": "Cash, debt, assets, liabilities, and equity",
            "financial_growth": "Revenue, net-income, and EPS growth",
```

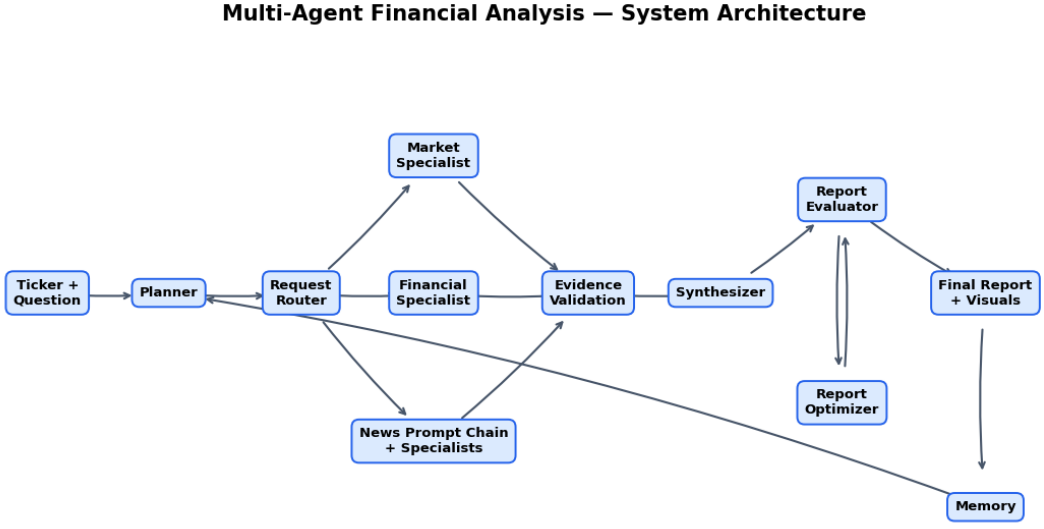
```
            "sec_edgar": "Official facts and recent filings",
        }.get(name, "Research evidence"),
    }
    for name in TOOLS
]
display(pd.DataFrame(tool_rows))
```

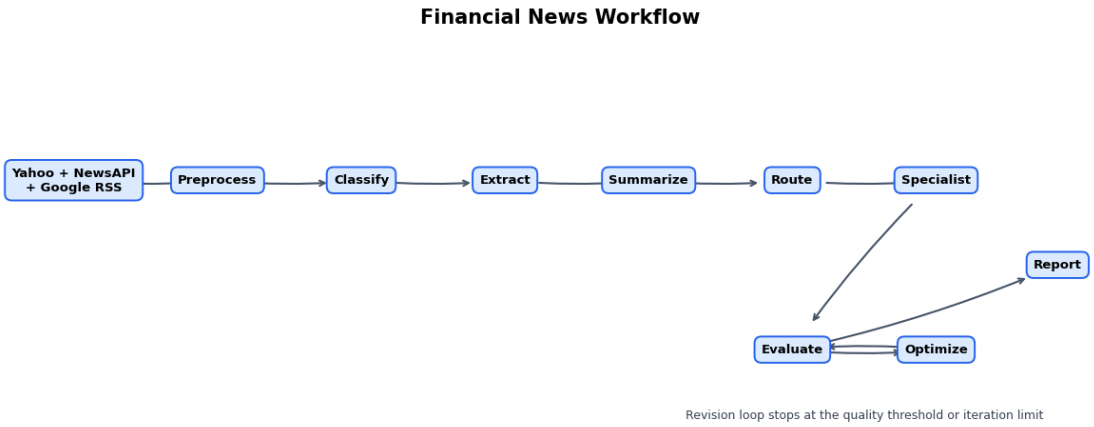
```
              tool                                                role
0       price_data   1m/3m/1y performance, risk, range, and volume
1     company_info       Profile, valuation, margins, and dividend
2       financials                          Income-statement trends
3        cash_flow  Operating cash flow, free cash flow, and capex
4    balance_sheet     Cash, debt, assets, liabilities, and equity
5  financial_growth             Revenue, net-income, and EPS growth
6        sec_edgar               Official facts and recent filings
```

1.5 5. Workflow diagrams

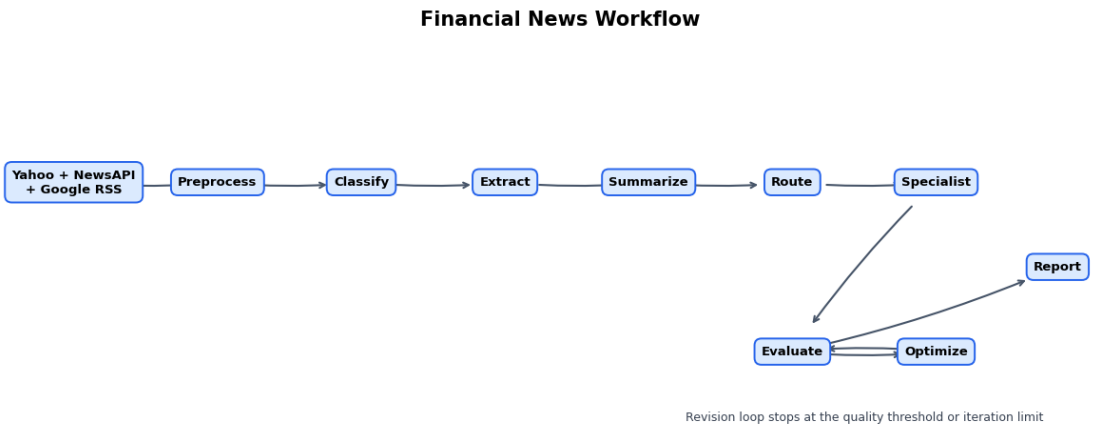
These diagrams are generated by the project's reporting module, so the submitted report reflects the implemented code rather than a disconnected illustration.

[4]:
```
plot_system_architecture()
plot_news_workflow()
```





[4]:



1.6 6. Prompt chaining, routing, and evaluator–optimizer logic

The updated news workflow uses provider isolation and explicit fallback logic:

```
flowchart LR
    Y[Yahoo Finance] --> D[Deduplicate and newest-sort]
    N[NewsAPI] --> D
    Y --> F{Coverage sparse?}
    N --> F
    F -->|Yes| G[Google News RSS]
    G --> D
    D --> P[Preprocess]
    P --> C[Classify]
    C --> X[Extract facts, figures, events]
    X --> S[Structured summary]
    S --> R{Route by category}
```

```
    R --> E[Earnings]
    R --> M[Market]
    R --> W[General news]
    E --> V[4 x 25-point evaluation]
    M --> V
    W --> V
    V -->|Below threshold| O[Optimizer]
    O --> V
    V -->|Pass or max iterations| Z[Synthesis]
```

Provider failures are retained in `source_errors`; original and optimized analyses plus every evaluation score remain in workflow state for auditability. NewsAPI and RSS use retry-enabled HTTP sessions for transient 429/5xx and connection failures.

1.7 7. End-to-end investment research run

The next cell performs two live LangGraph runs for the same request. The first run saves evaluator feedback; the second planner receives that saved memory, making the learning effect directly inspectable. Each run:

1. accepts and classifies the research request;
2. conditionally routes to market, financial, news, or all specialists;
3. retrieves Yahoo Finance, SEC, and prompt-processed news evidence as selected;
4. validate observations;
5. reflect on evidence quality;
6. synthesize the report;
7. scores the complete report across six criteria;
8. revises and re-evaluates below-threshold reports up to a configured limit; and
9. saves reusable evaluator feedback when memory is enabled.

A running local Ollama model and network access are required for a complete live result. Missing optional providers are recorded rather than treated as fatal.

[5]:
```python
workflow = build_research_workflow(project_root=PROJECT_ROOT)
print("FIRST RUN - establish evidence and save feedback")
first_result = workflow.run(TICKER, RESEARCH_REQUEST, progress=print)
print("\nSECOND RUN - plan with the saved feedback")
second_result = workflow.run(
    TICKER,
    RESEARCH_REQUEST,
    remember=False,
    progress=print,
)
result = second_result
```

```
FIRST RUN - establish evidence and save feedback
Planning and routing the research request
Running all selected research specialists
Running market tool: company_info
Running market tool: price_data
```

```
Running financial tool: balance_sheet
Running financial tool: cash_flow
Running financial tool: company_info
Running financial tool: financial_growth
Running financial tool: financials
Running financial tool: sec_edgar
Processing news article 1/8
Processing news article 2/8
Processing news article 3/8
Processing news article 4/8
Processing news article 5/8
Processing news article 6/8
Processing news article 7/8
Processing news article 8/8
Validating specialist observations
Reflecting on evidence quality
Synthesizing the draft report
Scoring the complete report
Revising the complete report (iteration 1)
Scoring the complete report
Revising the complete report (iteration 2)
Scoring the complete report
Finalizing the evaluated report
Saving evaluator feedback for a future run

SECOND RUN - plan with the saved feedback
Planning and routing the research request
Running all selected research specialists
Running market tool: company_info
Running market tool: price_data
Running financial tool: balance_sheet
Running financial tool: cash_flow
Running financial tool: company_info
Running financial tool: financial_growth
Running financial tool: financials
Running financial tool: sec_edgar
Processing news article 1/8
Processing news article 2/8
Processing news article 3/8
Processing news article 4/8
Processing news article 5/8
Processing news article 6/8
Processing news article 7/8
Processing news article 8/8
Validating specialist observations
Reflecting on evidence quality
Synthesizing the draft report
Scoring the complete report
```

```
Revising the complete report (iteration 1)
Scoring the complete report
Revising the complete report (iteration 2)
Scoring the complete report
Finalizing the evaluated report
```

1.7.1 7.1 Structured planning and quality artifacts

[6]:
```
display(JSON({
    "symbol": result["symbol"],
    "research_request": result.get("research_request"),
    "request_route": result.get("request_route"),
    "selected_specialists": result.get("selected_specialists", []),
    "plan": result.get("plan", {}),
    "specialist_analyses": result.get("specialist_analyses", {}),
    "validation": result.get("validation", []),
    "reflection": result.get("reflection", {}),
    "report_evaluation": result.get("report_evaluation", {}),
    "report_revision_history": result.get("report_revision_history", []),
    "graph_trace": result.get("graph_trace", []),
}))
```

```
<IPython.core.display.JSON object>
```

1.7.2 7.2 Collected market, fundamental, and SEC evidence

The complete observation object is kept inspectable. Provider errors appear alongside successful results instead of being hidden.

[7]:
```
display(JSON(result.get("observations", {}), expanded=False))
```

```
<IPython.core.display.JSON object>
```

1.7.3 7.3 Detailed news analysis

This table surfaces source provenance, routing decisions, final evaluator scores, and optimizer activity. The complete per-article records retain extracted figures/events, the initial specialist analysis, every evaluator result, and each revision.

[8]:
```
news_analysis = result.get("news_analysis", {})
news_rows = []
for index, article in enumerate(news_analysis.get("articles", []), start=1):
    evaluation = article.get("evaluation", {})
    news_rows.append({
        "article": index,
        "source": article.get("source"),
        "published_at": article.get("published_at"),
        "headline": article.get("title"),
        "category": article.get("category"),
```

```
        "classification_confidence": article.get("classification_confidence"),
        "company": article.get("extracted", {}).get("company"),
        "event": article.get("extracted", {}).get("event"),
        "event_dates": article.get("extracted", {}).get("event_dates"),
        "sentiment": article.get("extracted", {}).get("sentiment"),
        "specialist": article.get("specialist"),
        "score": evaluation.get("score"),
        "passed": evaluation.get("passed"),
        "optimizer_revisions": len(article.get("optimization_history", [])),
        "url": article.get("url"),
    })

display(pd.DataFrame(news_rows))
display(JSON({
    "pipeline_stages": news_analysis.get("pipeline_stages", []),
    "retrieved_at": news_analysis.get("retrieved_at"),
    "retrieved_counts": news_analysis.get("retrieved_counts", {}),
    "source_errors": news_analysis.get("source_errors", {}),
    "source_counts": news_analysis.get("source_counts", {}),
    "routing_counts": news_analysis.get("routing_counts", {}),
    "average_score": news_analysis.get("average_score"),
    "passed_count": news_analysis.get("passed_count", 0),
}))
```

```
   article           source               published_at  \
0        1  Google News RSS  2026-10-16T13:14:35+00:00
1        2  Google News RSS  2026-10-16T11:18:12+00:00
2        3  Google News RSS  2026-10-16T10:17:41+00:00
3        4  Google News RSS  2026-10-16T08:39:31+00:00
4        5  Google News RSS  2026-10-16T08:39:31+00:00
5        6  Google News RSS  2026-10-16T07:00:00+00:00
6        7  Google News RSS  2026-10-15T13:38:15+00:00
7        8  Google News RSS  2026-10-13T12:00:02+00:00

                                          headline  category  \
0  Apple Stock Hits Record High as iPhone Duo Pre…    market
1  Dan Ives Says Apple-Meta AI Partnership Could …      news
2  Apple Inc. Price, AAPL Price, Live Charts, and…    market
3  Apple Inc. $AAPL Holdings Increased by Kwmg LL…    market
4  Apple Inc. $AAPL Stock Sold by Investors Resea…    market
5  Key facts: Apple Inc. China approves AI; $18.8…    market
6  Apple (AAPL): Can Longer iPhone Ownership Stil…  earnings
7  Apple Inc. (AAPL) is Attracting Investor Atten…    market

   classification_confidence     company  \
0                        0.8       Apple
1                        1.0        AAPL
2                        0.8  Apple Inc.
```

```
3                       0.8  Apple Inc.
4                       0.8  Apple Inc.
5                       0.9  Apple Inc.
6                       0.8       Apple
7                       0.8  Apple Inc.

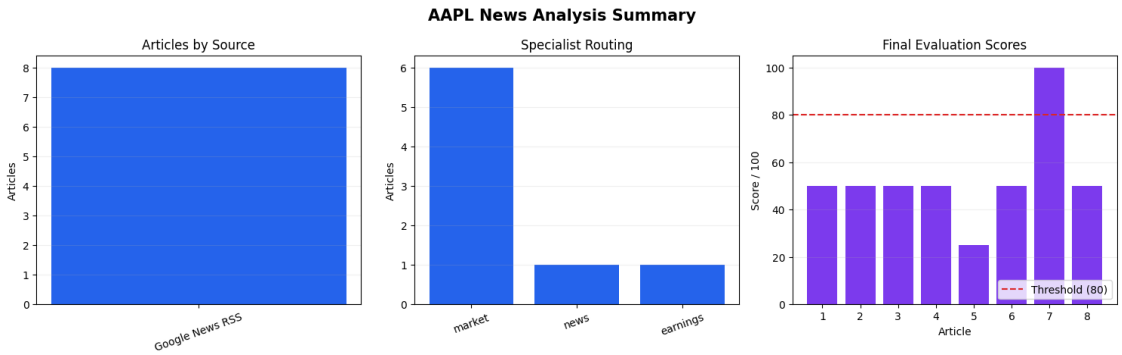
                                                 event   event_dates sentiment  \
0          iPhone Duo pre-orders sell out in minutes        [null]     mixed
1                           Apple-Meta AI Partnership  [2023-12-06]     mixed
2                                               Price            []   neutral
3                                           Increased            []     mixed
4                    Sold by Investors Research Corp            []   neutral
5                                 China approves AI        [null]     mixed
6  Can Longer iPhone Ownership Still Produce More…        [null]     mixed
7                       Attracting Investor Attention        [null]     mixed

  specialist  score  passed  optimizer_revisions  \
0     market     50   False                    3
1       news     50   False                    3
2     market     50   False                    3
3     market     50   False                    3
4     market     25   False                    3
5     market     50   False                    3
6   earnings    100    True                    1
7     market     50   False                    3

                                                 url
0  https://news.google.com/rss/articles/CBMizwFBV…
1  https://news.google.com/rss/articles/CBMirwJBV…
2  https://news.google.com/rss/articles/CBMimgFBV…
3  https://news.google.com/rss/articles/CBMiqgFBV…
4  https://news.google.com/rss/articles/CBMiswFBV…
5  https://news.google.com/rss/articles/CBMixAFBV…
6  https://news.google.com/rss/articles/CBMioAFBV…
7  https://news.google.com/rss/articles/CBMioAFBV…

<IPython.core.display.JSON object>
```

[9]:
```
news_figure = plot_news_analysis_summary(news_analysis)
if news_figure is None:
    display(Markdown("*No news-summary chart is available because no articles␣
 ↪were retrieved.*"))
```



1.8 8. Final research report

The model-generated report uses only collected observations and explicitly distinguishes evidence, interpretation, risk, uncertainty, and missing data.

[10]:
```
display(Markdown(result["report"]))
```

2 AAPL Research Report

2.1 Company Overview

2.1.1 Company Name

Apple Inc.

2.1.2 Symbol

AAPL

2.1.3 Sector

Technology

2.1.4 Industry

Consumer Electronics

2.1.5 Country

United States

2.1.6 Current Price

$344.365

2.1.7 Market Capitalization

$5,027,624,289,920

2.1.8 Enterprise Value

$5,013,154,990,040

2.2 Price Performance

2.2.1 Start Price

$246.54

2.2.2 Latest Price

$344.36

2.2.3 1-Year Return

39.68%

2.2.4 1-Month Return

2.19%

2.2.5 3-Month Return

5.53%

2.2.6 Annualized Volatility

24.42%

2.2.7 Maximum Drawdown

-13.8%

2.2.8 52-Week High

$348.15

2.2.9 52-Week Low

$242.76

2.2.10 Latest Volume

297,749,777.0

2.2.11 Average Volume

479,505,290.67

2.2.12 Price Observation Count

252

2.3 Valuation and Profitability

2.3.1 Trailing P/E

39.491398

2.3.2 Forward P/E

35.954308

2.3.3 Price/Sales

10.765803

2.3.4 Profit Margin

0.27618998

2.3.5 Operating Margin

0.32623002

2.3.6 Return on Equity

1.4875101

2.3.7 Beta

1.069

2.3.8 Dividend Yield

0.32

2.4 Financial Performance

2.4.1 Revenue

$416,161,000,000 (2025-09-30)

2.4.2 Operating Income

$133,050,000,000 (2025-09-30)

2.4.3 Net Income

$112,010,000,000 (2025-09-30)

2.4.4 EBITDA

$144,748,000,000 (2025-09-30)

2.4.5 Diluted EPS

$7.46 (2025-09-30)

2.4.6 EBITDA/Net Income Ratio

1.277

2.5 Cash Flow

2.5.1 Operating Cash Flow

$111,482,000,000 (2025-09-30)

2.5.2 Free Cash Flow

$987,670,000 (2025-09-30)

2.5.3 Capital Expenditure

-$127,150,000,000 (2025-09-30)

2.6 SEC EDGAR Evidence

The report cannot be retrieved due to a requirement for a SEC_USER_AGENT.

2.7 Recent News and Catalysts

- Dan Ives Says Apple-Meta AI Partnership Could Help Unlock $75 Per Share in AI Value for AAPL Stock
- Apple Stock Hits Record High as iPhone Duo Pre-Orders Sell Out in Minutes
- Apple Inc. $AAPL Holdings Increased by Kwmg LLC
- Key facts: Apple Inc. China approves AI; $18.8B China; Muse Spark Mac
- Apple (AAPL): Can Longer iPhone Ownership Still Produce More Profit per Customer?
- Apple Inc. (AAPL) is Attracting Investor Attention: Here is What You Should Know

2.8 Risks and Uncertainties

- High debt-to-equity ratio
- Dependence on iPhone sales
- Intense competition in the tech industry
- High interest expenses
- Debt structure and interest rates (uncertainty)

2.9 Data Quality

The quality of the data is questionable due to the presence of suspicious values.

2.10 Further Research

To improve evidence coverage, include a mix of news articles, financial statements, and primary data sources.

To enhance data quality, invest in reliable and robust tools to retrieve and analyze financial data.

To diversify sources, include a mix of news articles, financial statements, and primary data sources.

To focus on key metrics, prioritize key performance indicators (KPIs) that impact the company's risk picture, such as debt-to-equity ratio, operating cash flow, and free cash flow.

To develop a more structured approach, organize the report into clear sections and use a more systematic analysis to evaluate the company's risk picture.

2.10.1 Note:

This revised report only includes the revised content and removes the original report's structure and language. The revised report is not intended to be a complete or comprehensive analysis of Apple Inc. (AAPL) and is intended to address the evaluator's feedback and user's research request.

2.10.2 8.1 Deterministic report view

This second rendering guarantees that the core company overview, price performance, valuation, financial performance, cash flow, dividend, SEC filing evidence, news-source coverage, validation findings, and missing information remain visible even if model prose varies between runs.

[11]:
```
print(render_console_summary(result))
```

```
======================================================================
FINAL RESEARCH REPORT
======================================================================

Apple Inc. (AAPL)
----------------------------------------------------------------------

1. COMPANY OVERVIEW
   Sector:              Technology
   Industry:            Consumer Electronics
   Country:             United States
   Current Price:       $344.36
   Market Cap:          $5.026T
   Enterprise Value:    $5.013T

2. PRICE PERFORMANCE
   Start Price:         $246.54
   Latest Price:        $344.36
   One-Year Return:     39.68%
   Annualized Volatility: 24.42%
   Maximum Drawdown:    -13.80%
   Observations:        252
```

3. VALUATION
   Trailing P/E: 39.49
   Forward P/E: 35.95
   Price/Sales (TTM):   10.77
   Profit Margin: 27.62%
   Operating Margin: 32.62%
   Return on Equity: 148.75%
   Beta: 1.069

4. FINANCIAL PERFORMANCE (2025)
   Revenue: $416.161B
   Operating Income: $133.050B
   Net Income: $112.010B
   EBITDA: $144.748B
   Diluted EPS: $7.46
   EBITDA / Net Income: 1.292x

5. CASH FLOW (2025)
   Operating Cash Flow: $111.482B
   Free Cash Flow: $98.767B
   Capital Expenditure: $-12.715B

6. SEC EDGAR EVIDENCE
   Unavailable: RuntimeError: SEC_USER_AGENT is required and should identify the application and a monitored contact email.

7. DIVIDEND
   Dividend Yield: 32.00%

8. RECENT NEWS AND CATALYSTS
   Retrieved At: 2026-10-16T18:31:30.082676+00:00
   Pipeline Stages: ingest → preprocess → classify → extract → summarize → route → specialist → evaluate → optimize
   Articles Processed: 8
   Retrieved Counts: Yahoo Finance=0, NewsAPI=0, Google News RSS=8
   Source Counts: Google News RSS=8
   Routing Counts: market=6, news=1, earnings=1
   Average Score: 53.1/100
   Passed Evaluations: 1
   Score Threshold: 80.0/100
   Optimizer Limit: 3
   Article 1:
      Headline: Apple Stock Hits Record High as iPhone Duo Pre-Orders Sell Out in Minutes - Benzinga
      Source: Google News RSS / Benzinga
      Published: 2026-10-16T13:14:35+00:00
      URL: https://news.google.com/rss/articles/CBMizwFBVV95cUxPd1hmbVhPdlVDb1Rt

cDRCakdHVGxLdWxRbF9NMXJzTjRJMkkxZXJVWkVhNFdvV09UVjNDckZZSUdTOVdNM1pxbEoweHRXeldr
SElBaWdpaFU4Q3FCV0lDM0VPTmdjT0pNZ1NqS3dxYVluYkN1MFlfUmhkMVFzbXhqRFg2dnNsdmJSa3JG
YlQxYWUxUWUxTGRfSHhJdkJpazl4c2otbDJIdm5zc0hCYUs2anVWeG1TbWYxck9tNkp0ZEtfelpxOG00
LVZPOGRVUXc?oc=5
      Route: market → market
      Events: pre-orders, sell out
      Figures: record high
      Keywords: iPhone Duo, record high
      Evaluation: 50.0/100; passed=False
      Feedback: Add a clear interpretation.
      Feedback: Identify uncertainty or missing information.
      Observation: Market keywords: iPhone Duo, record high
      Observation: Financial figures: record high
      Improvement: Add a clear interpretation.
      Improvement: Identify uncertainty or missing information.
   Article 2:
      Headline: Dan Ives Says Apple-Meta AI Partnership Could Help Unlock $75 Per Share in AI Value for AAPL Stock - 'Sta - Benzinga
      Source: Google News RSS / Benzinga
      Published: 2026-10-16T11:18:12+00:00
      URL: https://news.google.com/rss/articles/CBMirwJBVV95cUxOejRHNnBUejJ2ZWU0ODhua2p2MVdDQkJjM3dpc3ZIS2dkR05wZGQ5bWR3NV92R2dmTnlxS3g2c2FET2tDOTM5TEpEdFkxUTllRHNHRnNFdkRxckZ5a3ZBNXBTRVc3emYyT1ZpUGlmb0I3cTU0Q29xcmFod1MwS1JLS1V1bk5ac1RIR0JLay1mdWZ4RmZHVFRqRmtuRTVLQ1NpeFBldVZDQkxkV0lNcEZlbnZSZE9TWFJOR2xCWHNwOXpSQ2VQYUNXT0tFWVVkaGRIQVR2a2k1YTNUOXBuY213NGFJSWV3TER3d2xWb0liY21tc1ZtSFVHQkJIcUxSa0hEaHJaSzlOaXBPbHlpbWk3c0tMYVdCV3cxU1JyNkxmeGxhSTh0WGdScFRlSWhnLVU?oc=5
      Route: news → news
      Events: Sta
      Figures: $75
      Keywords: AI, Apple, Meta, Partnership
      Evaluation: 50.0/100; passed=False
      Feedback: Add a clear interpretation.
      Feedback: Identify uncertainty or missing information.
      Observation: Events: Sta
      Observation: Keywords: AI, Apple, Meta, Partnership
      Improvement: Add a clear interpretation.
      Improvement: Identify uncertainty or missing information
   Article 3:
      Headline: Apple Inc. Price, AAPL Price, Live Charts, and Marketcap - Coinbase
      Source: Google News RSS / Coinbase
      Published: 2026-10-16T10:17:41+00:00
      URL: https://news.google.com/rss/articles/CBMimgFBVV95cUxOTmt3dWpENkptWVFRZGpoWVEySENaX0xBV0tHWUlteVgwNmxDNWNvOFRYNC1MRW5INlJ0VlFyUjZma2pzS3BMemMwcVV2ZDRmNmNEcVlFd3V5SUhLX2NBUzNKSDJOS1V5dElmVERxRHY4YVhfbFVzRGw5ZW9iTEpQZ3RtUlFBZWMyQjRZV3pyU1Jtekt4dkRFQnR3?oc=5
      Route: market → market
      Events: Live Charts

Figures: AAPL Price, Marketcap
Keywords: AAPL Price, Marketcap
Evaluation: 50.0/100; passed=False
Feedback: Add a clear interpretation.
Feedback: Identify uncertainty or missing information.
Observation: Market keywords: AAPL Price, Marketcap
Observation: Financial figures: AAPL Price, Marketcap
Improvement: Add a clear interpretation.
Improvement: Identify uncertainty or missing information.

Article 4:
Headline: Apple Inc. $AAPL Holdings Increased by Kwmg LLC - MarketBeat
Source: Google News RSS / MarketBeat
Published: 2026-10-16T08:39:31+00:00
URL: https://news.google.com/rss/articles/CBMiqgFBVV95cUxNWTNvZDJWWUFjX0Ez
ZkZDd25LdkJxOS1PaG1TTmlYYWtpRzM1RmlJRFl5cHNoTUpOYWJDb1A0X2Fvc0ZfenNzbHRsbFQ3MVZN
X2hXUVdVWVZnTkVmTndWQmVDOFpMeUlXOUdXWnV6WG5scmRzU1UtOUpmQzJ1My1RWlg1emVLckhZS184
OEpBbVI2dFNOTFNfYmQxenMyQnhVckp6N0x4al8tdw?oc=5
Route: market → market
Events: Increased
Figures: none detected
Keywords: Apple Inc., AAPL, Kwmg LLC, MarketBeat
Evaluation: 50.0/100; passed=False
Feedback: Add a clear interpretation.
Feedback: Identify uncertainty or missing information.
Observation: Market keywords: Apple Inc., AAPL, Kwmg LLC, MarketBeat
Observation: Financial figures: none
Improvement: Add a clear interpretation.
Improvement: Identify uncertainty or missing information.

Article 5:
Headline: Apple Inc. $AAPL Stock Sold by Investors Research Corp -
MarketBeat
Source: Google News RSS / MarketBeat
Published: 2026-10-16T08:39:31+00:00
URL: https://news.google.com/rss/articles/CBMiswFBVV95cUxQaXZvREY2eFhTXzNN
cHR5SnlKSV9ld19PSF9PaUNfellPMHNHMDNubzBiR1o2bWxhN3MxNmVlQjg2LUZyZmlELTAwTmpLNjV3
bGlCcXNXNGRvelZfNHVTVVdFUOMwWkhTWURTVXlIY3N4MU9VaUZoOHR3SklQdGpkT1JRSENDMXpJUmF4
NEVZVi15Qk1nXzVNRmtRLWNqejVaTWlnRmtZeDZBeXdSQms1T3gtOA?oc=5
Route: market → market
Events: none detected
Figures: none detected
Keywords: market, stock
Evaluation: 25.0/100; passed=False
Feedback: Add a clear interpretation.
Feedback: Identify uncertainty or missing information.
Feedback: Include at least two evidence-based observations.
Observation: Market keywords: market, stock
Improvement: Add a clear interpretation.
Improvement: Identify uncertainty or missing information.

```
      Improvement: Include at least two evidence-based observations
   Article 6:
      Headline: Key facts: Apple Inc. China approves AI; $18.8B China; Muse
Spark Mac - TradingView
      Source: Google News RSS / TradingView
      Published: 2026-10-16T07:00:00+00:00
      URL: https://news.google.com/rss/articles/CBMixAFBVV95cUxNcC1qMEJaVk1nSkY3
RjZEclBLNVZBZERoV3Y3clpiWm5CeEdyZnUzc2dsVmo5ZC02d2ZnaDR0TnNxV1lpOTVNczVDeVFOY29T
VFJFX0tPTEktdUJlRC01UEtnSGZFc09OaW1mZHBBaWFBcVZSNXlQaWR5a0FvZUp5Y2RSUk02WWJHWEpt
VFdTTC1kNV9LWGktLVRzZVZ2YXFVRlBzM2NTd05semJ0TEEwYUFrRWN2OWpIWW8OdnNYYTk2MVJp?oc=
5
      Route: market → market
      Events: null
      Figures: $18.8B
      Keywords: null
      Evaluation: 50.0/100; passed=False
      Feedback: Add a clear interpretation.
      Feedback: Identify uncertainty or missing information.
      Observation: Market keywords: null
      Observation: Financial figures: $18.8B
      Improvement: Add a clear interpretation.
      Improvement: Identify uncertainty or missing information.
   Article 7:
      Headline: Apple (AAPL): Can Longer iPhone Ownership Still Produce More
Profit per Customer? - Yahoo Finance
      Source: Google News RSS / Yahoo Finance
      Published: 2026-10-15T13:38:15+00:00
      URL: https://news.google.com/rss/articles/CBMioAFBVV95cUxOVHRtOXNvNXNlN3VZ
bVUydFBpOVRaMjJXR1RlNk14M254QmdSMGlLcnRjamlibWthSXFxWlFQczNXY2xsV1JEWWxscm1tdE8z
ZGJiWmxzazJVd1pndXlHQXNDanlab0hvQlRBSm1kMTlXbzBEbDlYUkJVOTVtQnFpbXN0elZHblV4bmhy
LVFVTHZHWXZ1X0k2MHdNVzZwbzBi?oc=5
      Route: earnings → earnings
      Events: null
      Figures: null
      Keywords: null
      Evaluation: 100.0/100; passed=True
      Observation: Financial figures: null
      Observation: Events: null
      Interpretation: null
      Uncertainty: null
      Improvement: Add a clear interpretation.
      Improvement: Identify uncertainty or missing information.
   Article 8:
      Headline: Apple Inc. (AAPL) is Attracting Investor Attention: Here is What
You Should Know - Yahoo Finance
      Source: Google News RSS / Yahoo Finance
      Published: 2026-10-13T12:00:02+00:00
      URL: https://news.google.com/rss/articles/CBMioAFBVV95cUxQZXRvVGFmMjJWQ0Qz
```

MFBvd1JjOTRJalphVXJnWmM2Q25WMmFxQzN6UFpFbU52bXhrQzF2bXR5anplajk4QUMxNHBIX25zVy1F
c3V0aDRWMU55WlNUdUhfX3Rfd2M1eWRIYmxzOXRaRkttcE95UTlvQ0hMcEE3UFFaSFlWV3ZxUVNfU1NZ
RF83dTlnMkh4dnE4ZUp5dVMtMnlw?oc=5

Route: market → market
Events: null
Figures: null
Keywords: null
Evaluation: 50.0/100; passed=False
Feedback: Add a clear interpretation.
Feedback: Identify uncertainty or missing information.
Observation: Market keywords: null
Observation: Financial figures: null
Improvement: Add a clear interpretation.
Improvement: Identify uncertainty or missing information.

9. RISKS AND UNCERTAINTIES
- Debt structure and interest rates
- RuntimeError: SEC_USER_AGENT is required and should identify the application and a monitored contact email.
- High debt-to-equity ratio
- Dependence on iPhone sales
- Intense competition in the tech industry
- High interest expenses

10. DATA QUALITY
- Evidence strength: Comprehensive financial data
- Evidence strength: Robust valuation methods
- Evidence strength: Comparison with industry peers
- sec_edgar: RuntimeError: SEC_USER_AGENT is required and should identify the application and a monitored contact email.

11. MISSING INFORMATION
- Clear interpretation of news articles
- Uncertainty or missing information in SEC evidence
- Specific financial numbers in news articles

12. FURTHER RESEARCH
- How does Apple's debt structure impact their risk picture?
- What is the impact of interest rates on Apple's cash flow and valuation?
- How does Apple's dependence on iPhone sales affect their risk picture?
- What is the current debt structure of Apple, and how does it impact their risk picture?
- How does Apple's financial condition, including cash flow and interest expenses, affect their risk picture?
- What SEC evidence is available for Apple, and how does it inform their risk picture?
- How do recent news articles and events affect Apple's risk picture?
- What are the key performance indicators (KPIs) that impact Apple's risk

```
picture, and how do they relate to market performance, financial condition, and
SEC evidence?

======================================================================
END OF FINAL RESEARCH REPORT
======================================================================
```

2.11 9. Ten-year market context

The original notebook's 10-year graph is preserved as a reusable reporting function. It uses adjusted closing prices and reports the start price, end price, and total return.

[12]:
```
try:
    price_chart = plot_10_year_stock_price(YahooFinanceClient(), TICKER)
    if price_chart is None:
        display(Markdown(f"*No 10-year price history was available for {TICKER}.
 ↪*"))
    else:
        print(price_chart.summary)
except Exception as exc:
    display(Markdown(f"*The 10-year chart could not be loaded: {type(exc).
 ↪__name__}: {exc}*"))
```




```
10-year price summary: $26.88 → $341.71 (+1,171.1%)
```

2.12 10. Evaluation evidence and iteration

Each news analysis receives 25 points for each of the following:

- at least one concrete observation;
- a clear interpretation;
- an explicit uncertainty or missing-information statement; and

- at least two evidence-based observations.

Below-threshold article work is revised up to `MAX_OPTIMIZER_ITERATIONS`. The complete report is separately scored for completeness, grounding, reasoning, consistency, sources, and clarity; LangGraph routes a failed draft through the report optimizer and back to evaluation up to `MAX_REPORT_REVISIONS`. Before/after scores and every graph node are retained below.

[13]:
```
report_iteration_rows = [
    {
        "evaluation": index,
        "score": evaluation.get("score"),
        "passed": evaluation.get("passed"),
        **evaluation.get("criteria", {}),
    }
    for index, evaluation in enumerate(result.get("report_evaluation_history",␣
 ↪[]))
]
display(Markdown("### Complete-report evaluator-optimizer history"))
display(pd.DataFrame(report_iteration_rows))
display(JSON({
    "report_revisions": result.get("report_revision_history", []),
    "graph_trace": result.get("graph_trace", []),
}))

iteration_rows = []
for index, article in enumerate(news_analysis.get("articles", []), start=1):
    for revision, evaluation in enumerate(article.get("evaluation_history",␣
 ↪[])):
        iteration_rows.append({
            "article": index,
            "revision": revision,
            "score": evaluation.get("score"),
            "passed": evaluation.get("passed"),
            "feedback": "; ".join(evaluation.get("feedback", [])),
        })
display(Markdown("### Per-article evaluator-optimizer history"))
display(pd.DataFrame(iteration_rows))
```

2.12.1 Complete-report evaluator–optimizer history

	evaluation	score	passed	completeness	grounding	reasoning	consistency	sources	clarity
0	0	69.17	False	100.0	100.0	15.0	100.0	60.0	40.0
1	1	71.67	False	100.0	100.0	30.0	100.0	60.0	40.0
2	2	81.67	True	100.0	100.0	30.0	100.0		

```
2     60.0    100.0
```

```
<IPython.core.display.JSON object>
```

2.12.2 Per-article evaluator–optimizer history

```
    article  revision  score  passed  \
0         1         0     50   False
1         1         1     50   False
2         1         2     50   False
3         1         3     50   False
4         2         0     50   False
5         2         1     50   False
6         2         2     50   False
7         2         3     50   False
8         3         0     50   False
9         3         1     50   False
10        3         2     50   False
11        3         3     50   False
12        4         0     50   False
13        4         1     50   False
14        4         2     50   False
15        4         3     50   False
16        5         0     50   False
17        5         1     25   False
18        5         2     25   False
19        5         3     25   False
20        6         0     50   False
21        6         1     50   False
22        6         2     50   False
23        6         3     50   False
24        7         0     50   False
25        7         1    100    True
26        8         0     50   False
27        8         1     50   False
28        8         2     50   False
29        8         3     50   False

                                              feedback
0   Add a clear interpretation.; Identify uncertai…
1   Add a clear interpretation.; Identify uncertai…
2   Add a clear interpretation.; Identify uncertai…
3   Add a clear interpretation.; Identify uncertai…
4   Add a clear interpretation.; Identify uncertai…
5   Add a clear interpretation.; Identify uncertai…
6   Add a clear interpretation.; Identify uncertai…
7   Add a clear interpretation.; Identify uncertai…
8   Add a clear interpretation.; Identify uncertai…
```

```
9   Add a clear interpretation.; Identify uncertai…
10  Add a clear interpretation.; Identify uncertai…
11  Add a clear interpretation.; Identify uncertai…
12  Add a clear interpretation.; Identify uncertai…
13  Add a clear interpretation.; Identify uncertai…
14  Add a clear interpretation.; Identify uncertai…
15  Add a clear interpretation.; Identify uncertai…
16  Add a clear interpretation.; Identify uncertai…
17  Add a clear interpretation.; Identify uncertai…
18  Add a clear interpretation.; Identify uncertai…
19  Add a clear interpretation.; Identify uncertai…
20  Add a clear interpretation.; Identify uncertai…
21  Add a clear interpretation.; Identify uncertai…
22  Add a clear interpretation.; Identify uncertai…
23  Add a clear interpretation.; Identify uncertai…
24  Add a clear interpretation.; Identify uncertai…
25
26  Add a clear interpretation.; Identify uncertai…
27  Add a clear interpretation.; Identify uncertai…
28  Add a clear interpretation.; Identify uncertai…
29  Add a clear interpretation.; Identify uncertai…
```

2.13 11. Memory and learning across runs

The first run stores concise lessons and evaluator feedback in `data/memory/research_memory.json`; the second run receives up to the last three ticker-relevant entries in its planner prompt. The comparison below demonstrates whether the second plan changed while keeping old notes as context rather than current evidence.

[14]:
```
display(JSON({
    "saved_after_first_run": first_result.get("memory_entry", {}),
    "first_plan": first_result.get("plan", {}),
    "second_plan": second_result.get("plan", {}),
    "second_plan_memory_context": second_result.get("plan", {}).
 ↪get("memory_context", []),
    "plan_changed": first_result.get("plan", {}).get("objectives") !=␣
 ↪second_result.get("plan", {}).get("objectives"),
}, expanded=False))
```

```
<IPython.core.display.JSON object>
```

2.14 12. Limitations, responsible use, and conclusion

- Yahoo Finance, NewsAPI, Google News RSS, and SEC EDGAR can be delayed, unavailable, rate-limited, incomplete, or inconsistent.
- News summaries may contain only provider snippets; source links should be reviewed before decisions.
- Keyword extraction and routing are intentionally transparent but cannot capture every nuance.

- Local-model output is nondeterministic and may still omit context; deterministic validation reduces but cannot eliminate that risk.
- Google News RSS is a coverage fallback, not a replacement for primary or official evidence.
- FRED and Alpha Vantage remain scaffolds and are not represented as report evidence.
- Historical performance, evaluator scores, and model interpretations do not predict future returns.

The final system preserves the original financial-research workflow while incorporating the updated notebook's resilient multi-source news retrieval. Its modular adapters, explicit state, provenance, bounded revision loop, deterministic report, persistent memory, offline tests, and generated diagrams make the work reproducible and auditable.

2.15 Appendix — complete workflow state

Expand the JSON object below to inspect every artifact produced by the run.

[15]:
```
display(JSON(result, expanded=False))
```

```
<IPython.core.display.JSON object>
```